ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

PURE MATHEMATICS **6042/2**
PAPER 2 3 hours

SPECIMEN PAPER

Additional materials:
Answer booklet
List of Formulae MF 7
Scientific calculator [Non-programmable]

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided on the answer booklet.

Answer **all** questions in Section A and any **five** questions from Section B.

If a numerical answer cannot be given exactly and the accuracy required is not specified in the question, then in the case of an angle it should be given correct to the nearest degree, and in other cases it should be given correct to 2 significant figures.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

This specimen paper consists of 9 printed pages.

Section A [40 marks]
Answer **all** questions in this section.

1. Solve the following inequality $|5x - 7| > 2x + 7$. [4]

2. **(a)** **(i)** A sequence S_n is such that $U_{n+1} = 3U_n - 1$ and $U_1 = 1$

Find the first 5 terms of the sequence. [2]

(ii) Another sequence R_n is such that $P_{n+1} = 2S_n - 1$.

Find the first 4 terms of R_n. [2]

(iii) Determine the type of progression formed by the terms of R_n and hence state the formula for the n^{th} term. [2]

(iv) Calculate the sum of the first 8 terms of R_n. [2]

3. **(a)** Solve the equation

$$\sqrt{(1+x)^3} = 8$$ [2]

(b) **(i)** A drum has a cylindrical surface area Am^2 that varies jointly as its base diameter, *d*m and its height, *h* meters. Given that the surface area of the drum is 3.26m when its base radius is 0.97m and its height is 1.82m.

find the relationship between *A*, *d* and *h*. [3]

(ii) Calculate the radius of a cylinder whose surface area is 15.14m^2 and its height is 3.62m to 2 decimal places. [3]

4. (a) Solve the following equation

$3\left(e^{4x}\right) - 7\left(e^{2x}\right) + 2 = 0$, giving your answer in exact form. [5]

(b) Solve the differential equation

$$\frac{dy}{dx} = xy - 2x$$

expressing y in terms of x given that $y = 4$ when $x = 0$. [6]

5. (a) Express

$$\frac{4x^2 + 16x + 19}{x^2 + 4x + 4}$$

in the form

$$\frac{A}{(x+B)^2} + C$$

$x \neq -B$ where A, B and C are rational numbers. [2]

(b) State the sequence of transformations by which the graph of

$$y = \frac{4x^2 + 16x + 19}{x^2 + 4x + 4},$$

may be obtained from the graph

$$y = \frac{1}{x^2}.$$ [3]

(c) Evaluate

$$\int_0^1 \frac{4x^2 + 16x + 19}{x^2 + 4x + 4}\, dx.$$ [4]

Section B (80 marks)

Answer any **five** questions from this section. Each question carries **16** marks.

6. **(a)** Using Taylor series, find the first 5 terms of the function,

$$f(x) = (x-2)\,e^{2x},$$

near $x = 2$. [5]

(b) Given that

$$y = \frac{1}{\sqrt{1+5x} + \sqrt{1+2x}}$$

where $x > -\frac{1}{5}$ and $x \neq 0$,

show that $y = \frac{1}{3x}\left(\sqrt{1+5x} - \sqrt{1+2x}\right)$. [3]

(c) **(i)** Using the form

$$y = \frac{1}{3x}\left(\sqrt{1+5x} - \sqrt{1+2x}\right),$$

express y as a series in ascending powers of x up to and including the term in x^2. [5]

(ii) Hence by putting $x = \frac{1}{25}$, show that

$$\frac{5}{\sqrt{30} + \sqrt{27}} \approx \frac{4\ 689}{10\ 000}.$$ [3]

7. (a) Simplify

$\dfrac{(2+5i)^6}{(5-2i)^5}$, leaving the answer in the form $a+bi$

where $a\ ,b \in \mathbb{R}$. [3]

(b) Solve the equation

$z^5 - 1 + \sqrt{3}i = 0$,

giving the answer in the form

$r(cos\theta + i \sin\theta)$. [6]

(c) (i) Use DeMoivre's Theorem to express $\cos 5\theta$ in powers of $\cos\theta$ only. [5]

(ii) Hence or otherwise find the derivative of

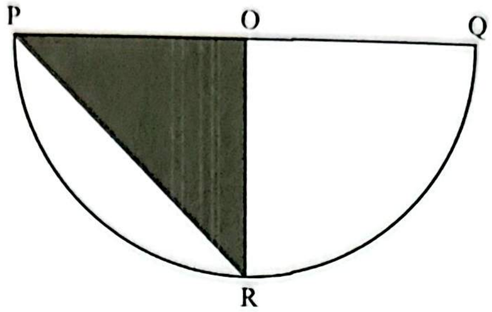
$16\,cos^5x - 20cos^3x$ [2]

8. (a) Starting with $x_0 = 3$, use the Newton Raphson method twice to find the root of

$x^3 - 3x^2 - 1 = 0$,

correct to three decimal places. [4]

(b) The diagram above shows a semi-circle PQR centre O and diameter PQ.



The point R on the semicircle is such that the area of the sector ROQ is equal to twice the area of the unshaded segment PR. Given that the radius is θ radians show that

$$3\theta = 2\left(\pi - Sin\theta\right).$$

[4]

(c) Let G be the set of all 3×3 matrices in the form

$$\begin{pmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{pmatrix}$$

where a, b, c $\in \mathbb{Z}^+$.

Prove that G forms a group under matrix multiplication. [8]

9. **(a)** **(i)** It is given that matrices

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix} \text{ and } A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

Describe the transformations represented by matrix A and matrix B. [2]

(ii) A point $(2; 3)$ is transformed by matrix A followed by matrix B. Write down the composite matrix of the transformations. [2]

(iii) Find the image of point A under the transformation in **(i)**. [1]

(b) **(i)** It is given that matrix

$$M = \begin{pmatrix} 2 & 6 & 1 \\ -1 & 2 & -1 \\ 4 & 3 & 1 \end{pmatrix}.$$

Find the inverse of matrix M. [7]

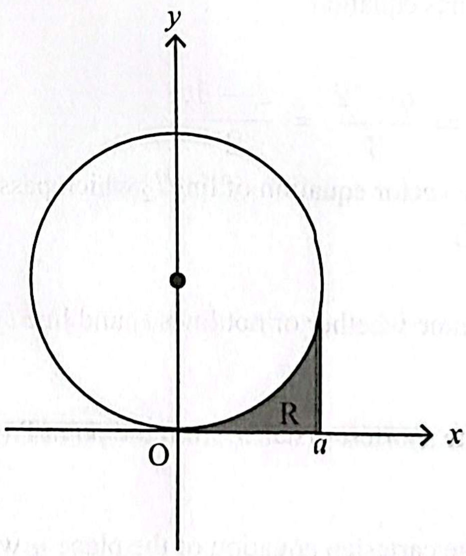
(ii) Hence or otherwise solve the following simultaneous equations

$$2x + 6y + x = 0$$
$$-x + 2y - z - 10 = 0$$
$$4x + 3y + z - 1 = 0$$

[4]

10. The diagram shows a sketch of a circle with equation

$x^2 + y^2 - 4y = 0.$



The shaded region R is bounded by the circle, $x - axis$ and line $x = a$

(a) Find the radius and centre of the circle

$x^2 + y^2 - 4y = 0.$ [2]

[Turn over

(b) **(i)** Express the equation of the circle in the form,

$$y = a - \sqrt{b - x^2}$$

where a and b are real numbers. [2]

(ii) Using substitution

$x = 2Sin\theta$ or otherwise,

find the area of shaded region R giving the answer in exact form. [7]

(iii) Find the volume of the solid generated when shaded region R is rotated through 2π radians about the $x - axis$. [5]

11. **(a)** **(i)** Points A and B have positions vectors

$3\boldsymbol{i} - 4\boldsymbol{k}$ *and* $2\boldsymbol{i} - \mathbf{j} + 3\boldsymbol{k}$ respectively.

Line l_1 has equation

$$\frac{x-2}{4} = \frac{y-2}{3} = \frac{z-1.}{2}$$

Find the vector equation of line l_2 which passes through points A and B. [2]

(ii) Determine whether or not lines l_1 and line l_2 intersect. [4]

(b) **(i)** Find the shortest distance from the point P (2;1;2) to the line l_2. [5]

(ii) Find the cartesian equation of the plane π which contains the line l_2 and point P (2;1;2). [5]

12. **(a)** **(i)** Solve the equation $3^{2x} = 4^{1-x}$, giving the answer in the form $\dfrac{\ln a}{\ln b}$ where $a, b \in \mathbb{Z}^{+}$. [3]

(ii) A linear graph of y against $\dfrac{1}{x^2}$ is drawn for the function

$$x^2(y-a) = b.$$

Find the values of a and b given that the graph passes through the points $(1; 4)$ and $(3; 8)$. [5]

(b) Prove by Induction that

$$\frac{1}{4} + \frac{1}{4^2} + \cdots + \frac{1}{4^n} = \frac{1}{3}\left(1 - \frac{1}{4^n}\right)$$

for $n \in \mathbb{Z}^{+}$. [8]